Roll. No.: _____

Thapar Institute of Engineering and Technology, Patiala
Department of Computer Science and Engineering
MST (Summer Semester), June 2026

Course Name: **Quantum Computing** | Course Code: **UCS619**
Total Marks: **30** | Faculty: **NEG**
Total Time: **120 Minutes** | Date: **June, 2026**

Note: Attempt all questions. Subparts of each question must be solved in one place.

1. (a) Two single-qubit systems are prepared in the following states: **(1+2+1)**

$$|\psi\rangle = \frac{|0\rangle + \sqrt{2}i\,|1\rangle}{\sqrt{3}}$$

$$|\phi\rangle = \frac{\sqrt{2}\,|0\rangle + i\,|1\rangle}{\sqrt{3}}$$

1. Verify whether both $|\psi\rangle$ and $|\phi\rangle$ are valid, normalized qubit states. Show all calculations.
2. Determine whether the two states $|\psi\rangle$ and $|\phi\rangle$ are orthogonal to each other.
3. Express the state $|\psi\rangle$ in the diagonal basis

$$\{|+\rangle, |-\rangle\}.$$

(b) Consider the two-qubit state **(2)**

$$|\psi\rangle = \frac{|01\rangle + |10\rangle}{\sqrt{2}}.$$

Find the resulting state after applying the following operator to $|\psi\rangle$.

$$Y \otimes Z$$

2. (a) A qubit is in the state **(3)**

$$|\psi\rangle = \frac{1}{\sqrt{3}}|0\rangle + \sqrt{\frac{2}{3}}|1\rangle$$

Using the projection operators P_0 and P_1, indicate the probability of finding $|0\rangle$ and the probability of finding $|1\rangle$ when a measurement is made.

(b) Find the tensor product of the Pauli matrices X and Z . **(2)**

(c) State the First Postulate of Quantum Mechanics. **(1)**

3. (a) Consider a two-qubit quantum circuit where both qubits start in the state $|00\rangle$. The circuit applies a Hadamard (H) gate to the first qubit (q_0), followed by a Controlled-NOT (CNOT) gate using q_0 as the control and q_1 as the target. **(1+1+1)**

(i) Write out the state of the system immediately after the Hadamard gate acts on q_0, but before the CNOT gate acts.

(ii) Show step-by-step how the CNOT gate modifies the state to produce the final output.

(iii) What is the common name given to this specific final two-qubit entangled state?

(b) Show that the Toffoli gate is its own inverse. (3)

4. (a) Alice wishes to transmit an unknown qubit

$$|\psi\rangle = \alpha\,|0\rangle + \beta\,|1\rangle\,,$$

to Bob using the quantum teleportation protocol. Alice and Bob share the entangled Bell state

$$\frac{|01\rangle + |10\rangle}{\sqrt{2}}.$$

Demonstrate the complete teleportation procedure. (4)

(b) Define No Cloning theorem and prove it. (2)

5. (a) Consider the Bell state

$$|\beta_{00}\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}.$$

(5)

In the Superdense Coding protocol, Alice applies one of the quantum gates I, X, Z, or iY to her qubit before sending it to Bob.

1. Determine the resulting Bell state when Alice applies the identity gate I.
2. Determine the resulting Bell state when Alice applies the Pauli-X gate.
3. Determine the resulting Bell state when Alice applies the Pauli-Z gate.
4. Determine the resulting Bell state when Alice applies the iY gate.
5. For each case above, identify the corresponding classical two-bit message (00, 01, 10, or 11) transmitted to Bob.

(b) Comment: Superdense Coding allows the transmission of two classical bits using one qubit and a shared entangled pair. (1)

End of the Question Paper

Question-wise COs and Bloom's taxonomy

Q.1	Q.2	Q.3	Q.4	Q.5
CO1, L2	CO1, L2	CO4, L3	CO2, L4	CO1, L3